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RESEARCH ARTICLE

NeuroFusion-Trans: A Novel Transformer-Based EEG-EMG Fusion Model for Assistive Robotics

TIPU SULTAN¹, GUANGPING LIU¹, (Student Member, IEEE),
PASCAL SIKORSKI², (Student Member, IEEE), SAMAH ALSHATHRI³,
WALID EL-SHAFAI^{4,5}, (Senior Member, IEEE), AND MADI BABAIASL¹, (Member, IEEE)

¹Aerospace and Mechanical Engineering Department, Saint Louis University, St. Louis, MO 63103, USA

²Computer Science Department, Saint Louis University, St. Louis, MO 63103, USA

³Department of Information Technology, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia

⁴Automated Systems and Computing Laboratory (ASCL), Computer Science Department, Prince Sultan University, Riyadh 11586, Saudi Arabia

⁵Department of Electronics and Electrical Communications Engineering, Faculty of Electronic Engineering, Menoufia University, Menouf 32952, Egypt

Corresponding author: Madi Babaiasl (madi.babaiasl@slu.edu)

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ABSTRACT User intent recognition from multimodal neurophysiological signals, particularly electroencephalography (EEG) and electromyography (EMG), is critical for enhancing human-machine interaction in assistive robotics. Recent advances in neurophysiological signal processing have enabled enhanced user intent recognition for assistive robotics and human-machine interfaces. However, achieving high accuracy and real-time adaptability in electromyography (EMG) and electroencephalography (EEG)-based gesture recognition remains challenging due to temporal misalignment, weak cross-modality fusion, and lack of adaptive learning. This paper proposes NeuroFusion-Trans, a novel transformer-based framework that improves EEG-EMG gesture recognition by improving temporal resolution, using cross-modality attention, and integrating adaptive online learning. Temporal resolution enhancement ensures dynamic EEG-EMG synchronization for improved signal alignment. The cross-modality attention mechanism captures interdependencies between EEG and EMG signals, leading to more accurate intent classification. Adaptive online learning enables real-time personalization by dynamically adjusting to user-specific variations. The model is evaluated on two publicly available EEG-EMG upper-limb gesture datasets: Dataset 1 (5,296 for training, 1,324 for validation) and Dataset 2 (5,276 for training, 1,304 for validation). NeuroFusion-Trans achieves state-of-the-art performance, with an accuracy of 97% and 96% and Cohen's Kappa of 0.97 and 0.95 after online adaptation, significantly outperforming baseline models such as CNN-LSTM, GRU, and LSTMNet. Ablation studies reveal that removing the cross-modality attention mechanism reduces accuracy by 6.1%, underscoring its importance in exploiting the EEG-EMG dependencies. Turning off synchronization leads to a 6.7% performance drop, demonstrating the necessity of real-time learning for robust intent recognition. Furthermore, NeuroFusion-Trans enhances EEG-EMG synchronization, achieving synchronization scores of 0.7471 on Dataset 1 and 0.7687 on Dataset 2, confirming its effectiveness in temporal alignment. The proposed approach demonstrates robust generalization, improved responsiveness, and potential applications in real-time assistive robotics. The implementation details and source code can be found at: <https://github.com/madibabaiasl/NeuroFusion-Trans>

INDEX TERMS Transformer, cross-modality attention, online learning, gesture recognition, assistive robotics, temporal synchronization.

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I. INTRODUCTION

Neural interfaces, particularly those leveraging EEG and EMG signals, have emerged as powerful tools for intuitive

human-machine interaction in applications such as assistive robotics and neurorehabilitation [1], [2]. Electroencephalography (EEG) and electromyography (EMG) are widely used biosignals for decoding human intent, offering non-invasive and real-time access to motor-related neural activity. EEG captures brain activity, reflecting cortical motor commands, while EMG records muscle activation patterns that directly correspond to movement execution. The integration of EEG and EMG signals has been explored to enhance gesture recognition accuracy and responsiveness [3], [4]. However, several challenges remain to exploit these signals effectively for real-time applications.

One major challenge is the temporal misalignment between EEG and EMG signals due to inherent differences in their signal generation processes and sampling rates [5]. EEG signals, originating from cortical activity, exhibit higher latency in reflecting motor intent compared to EMG signals, which capture muscle activation with minimal delay [5], [6]. This discrepancy can lead to synchronization errors, reducing the accuracy of intent recognition in time-sensitive applications. Another critical limitation is the lack of efficient cross-modality fusion between EEG and EMG signals. Traditional models often treat these modalities independently or rely on simple concatenation strategies, failing to exploit their complementary nature, resulting in suboptimal intent classification, particularly in complex motor tasks [7]. Personalized and adaptive learning remains an unresolved issue in EEG-EMG-based gesture recognition [8]. Inter-subject variability and non-stationarity in neural and muscular signals necessitate models that dynamically adapt to individual users over time [8], [9]. Most existing approaches rely on static training paradigms, limiting their generalizability to diverse users and physiological conditions. This lack of adaptability hinders the deployment of EEG-EMG interfaces in real-world applications, where continuous learning and user-specific adjustments are essential for reliable performance [10].

This paper introduces NeuroFusion-Trans, a transformer-based model that enhances EEG-EMG gesture recognition through dynamic synchronization, cross-modal feature fusion, and real-time adaptation. NeuroFusion-Trans addresses EEG-EMG misalignment by employing dynamic synchronization techniques. The cross-modality attention mechanism effectively fuses EEG and EMG information in a shared latent space, capturing interdependencies between brain and muscle activity to improve classification accuracy [3], [11]. Furthermore, the framework incorporates adaptive online learning, allowing user-specific continuous adjustments for enhanced robustness in real-world applications [12]. The key contributions of this paper are as follows.

- We propose NeuroFusion-Trans, a transformer-based framework for EEG-EMG gesture recognition, integrating temporal synchronization, cross-modality attention, and adaptive online learning.
- Temporal synchronization dynamically aligns EEG-EMG signals, reducing latency differences, while

cross-modality attention enhances feature fusion for improved classification.

- Adaptive online learning enables real-time personalization, improving model robustness across users and physiological variations.
- The model achieves state-of-the-art performance on two EEG-EMG datasets, with accuracies of 97% and 96%, and F1-scores of 0.97 and 0.96. Ablation studies reveal a 6.1% accuracy drop without cross-modality attention and a 6.7% drop without Synchronization for datasets. Synchronization scores of 0.7471 and 0.7687 further confirm the effectiveness of temporal alignment.

The remaining sections of the manuscript are as follows. Section II introduces the related literature analysis. Section III presents our novel proposed network for driver behavior detection. In Section IV, the results obtained from the implementation of various ML techniques are comparatively assessed. Section V summarizes the findings of this research study.

II. LITERATURE REVIEW

Recent advancements in deep learning and multimodal signal processing have significantly improved assistive robotics and neurorehabilitation intent recognition. However, existing methods often have limitations in cross-modality fusion, online adaptation, and real-time user feedback integration. This section reviews the most relevant works in EEG-EMG-based user intent recognition, highlighting their strengths and shortcomings.

A. MULTIMODAL LEARNING FOR INTENT RECOGNITION

Due to their complementary nature, EEG and EMG have been widely used for upper-limb gesture recognition [13]. Convolutional neural networks (CNN) and recurrent neural networks (RNN), such as long-short-term memory (LSTM) and gated recurrent units (GRU), have been used for time series modeling of EEG and EMG data [14]. For instance, proposed ShallowConvNet, which has been widely used in EEG-based classification due to its effectiveness in learning spatial and spectral EEG features [15]. However, unlike more recent transformer-based or attention-driven models, it does not incorporate explicit cross-modality learning strategies, making it less suited for EEG-EMG fusion.

Similarly, LSTMNet and GRUNet have been utilized to capture temporal dependencies in EEG signals, demonstrating effectiveness in learning long-term patterns, but challenges remain in achieving optimal performance due to the complexity of EEG's temporal dynamics [16], [17]. Transformer-based approaches, such as the TCN-Model [18], have been used for EEG-based motor imagery classification, improving long-range dependencies in EEG signals. However, none of these approaches have been extended to EEG-EMG fusion and fail to address adaptability and fine-grained cross-modal interactions of users.

B. CROSS-MODALITY ATTENTION MECHANISMS

A key limitation of traditional multimodal models is their inability to dynamically address the most relevant signal features across EEG and EMG channels. Cross-modal attention mechanisms have been proposed to address this challenge. Recent works, such as ChronoNet [19], integrate convolutional and recurrent layers to learn hierarchical features from EEG data. However, these models lack explicit attention mechanisms to weigh the importance of multimodal signals dynamically. Decisions for dynamic signal use in multimodal systems provide consideration for users who may be unable to provide consistent or accurate input in practice. In contrast, approaches like Mutual-Cross-Attention (MCA) for EEG feature fusion [20] and Self-Attention Multi-Temporal Convolutional Networks (SAMTCN) for EEG-EMG integration [21] have demonstrated the benefits of attention mechanisms in enhancing feature interactions and improving classification accuracy. They do not continuously incorporate online adaptation to refine user-specific intent patterns. Our proposed model, NeuroFusion-Trans, introduces an explicit cross-modality attention mechanism, allowing dynamic fusion of EEG and EMG signals for improved intent recognition. Prior work has demonstrated the effectiveness of transformer architectures in classifying real vs. imaginary motor intent using EEG data alone [22].

Our proposed model, NeuroFusion-Trans, introduces an explicit cross-modality attention mechanism, allowing dynamic fusion of EEG and EMG signals for improved intent recognition. The EMG-TransNN-MHA model introduced leverages multi-head attention mechanisms to address long-range dependencies in EMG data for motor intent recognition, achieving significant accuracy improvements [23]. However, it focuses solely on EMG data and does not incorporate multimodal integration with EEG signals or real-time adaptation, which are critical for universal assistive robotics applications. Our proposed NeuroFusion-Trans addresses these gaps by integrating cross-modality attention mechanisms and online adaptive learning.

C. ONLINE ADAPTATION AND INTERACTIVE FEEDBACK

Traditional deep learning models for gesture recognition are based on static training data that limits their ability to adapt to real-time variations in user signals. Some recent approaches, such as adaptive learning frameworks in Brain-Computer Interfaces (BCIs) [24], have incorporated incremental learning techniques but often require significant computational resources. Few studies have explored real-time adaptation in EEG-EMG-based models. Variational Autoencoder (VAE)-based approaches [25] have attempted to learn generative latent spaces for continuous adaptation but do not explicitly incorporate user feedback during real-time execution. Our approach introduces an online adaptive learning mechanism that fine-tunes the model based on real-time user intent corrections, enhancing long-term usability for assistive robotics applications.

Our approach integrates an online adaptive learning mechanism that fine-tunes the model in real time based on user intent corrections, ensuring long-term usability for assistive robotics. A cyclic learning rate schedule and replay stabilize adaptation while gradient clipping and regularization prevent catastrophic forgetting, preserving learned patterns. Our EEG-EMG Transformer maintains high classification performance in non-stationary conditions by dynamically updating model weights, which is crucial for neurorehabilitation where intent patterns evolve. Cross-modality attention enhances robustness to signal noise and inter-user variability, enabling assistive robotics to adapt reliably to users' changing needs.

D. LIMITATIONS OF EXISTING MODELS

While CNN-LSTM, GRU, and LSTM architectures have achieved success in unimodal time series modeling, they present major limitations for multimodal EEG-EMG fusion. Traditional recurrent models struggle with temporal misalignment between modalities due to inherent latency differences, and lack mechanisms for dynamic synchronization [5], [6]. CNN-LSTM frameworks typically concatenate features without explicitly modeling interdependencies, limiting the fusion quality [7]. Moreover, these models lack cross-modality attention mechanisms that adaptively weigh EEG and EMG contributions based on task relevance [20]. Static offline training further restricts adaptability to user-specific or session-specific variations [8]. Finally, sequential processing in RNN-based models such as LSTM and GRU introduces higher inference latency, making them less suitable for real-time assistive applications [18]. These challenges motivate the development of NeuroFusion-Trans, which addresses synchronization, fusion, and online adaptability for robust real-time performance.

E. RESEARCH GAP AND QUESTIONS

Despite progress in EEG-EMG-based gesture recognition, existing models struggle with temporal misalignment, inadequate cross-modal fusion, and limited real-time adaptability. Traditional deep learning architectures fail to synchronize EEG and EMG signals effectively, leading to suboptimal performance in real-time applications. Additionally, current methods lack an adaptive learning mechanism to enhance robustness across users and sessions. Our study addresses the following questions:

- Can a transformer-based framework with high-temporal-resolution synchronization improve EEG-EMG intent recognition over existing CNN, LSTM, and hybrid models?
- How does cross-modality attention impact the fusion of EEG and EMG signals, and can it enhance classification accuracy by dynamically weighting their contributions?
- Can an online adaptive learning mechanism improve model generalization and robustness across user conditions and signal variations?

We propose NeuroFusion-Trans, a transformer-based framework that integrates temporal synchronization, cross-modality attention, and online adaptation to address these challenges. By dynamically fine-tuning model parameters and improving multimodal integration, our approach enhances classification accuracy, real-time adaptability, and robustness, making it well-suited for assistive robotics and neurorehabilitation applications.

III. PROPOSED METHODOLOGY

The NeuroFusion-Trans framework is a transformer-based model designed for upper-limb gesture classification using EEG and EMG signals. The model incorporates three key innovations: temporal resolution enhancement, cross-modality attention, and adaptive learning. These components collectively improve gesture recognition accuracy, synchronization across EEG-EMG modalities, and real-time adaptability for assistive robotics applications.

A. DATASET DESCRIPTION

Two datasets were used for training and evaluation. Dataset 1, the EMG-EEG dataset for upper-limb gesture classification [26], and Dataset 2, the 8-channel EMG-EEG upper-limb gesture data [27]. The details of the data sets are presented in Table 1. The data was collected using a defunct Myo armband consisting of 8 channels with a sampling frequency of 250 Hz. All hand gestures performed by the participants are labeled from G1 to G7, and descriptions of each gesture are showcased in Table 2.

As shown in Table 1, both datasets involve EEG and EMG recordings of upper-limb gestures but differ slightly in complexity, environment, and demographics. Dataset 1 includes seven distinct gestures performed in seated conditions by 33 people with no known disabilities, with gestures such as prismatic pinch and cut grasp involving complex muscle coordination. Dataset 2 also includes seven gestures but adds a motor imagery (MI) component and includes two repetitions performed standing, introducing slight environmental and biomechanical variability. The Dataset 2 cohort consists of 11 subjects (7 males, 3 females, 1 left-handed male; mean age 25 ± 5.5 years), offering demographic specificity but a smaller sample size, which may limit generalization. These differences were considered during model training and evaluation, and potential impacts such as signal variability due to postural changes and cognitive load were mitigated through preprocessing and online adaptation mechanisms.

B. DATA CLEANING AND PREPROCESSING

A multi-stage data preprocessing pipeline was implemented to ensure high-quality EEG and EMG signals for gesture recognition, including noise removal, bandpass filtering, normalization, segmentation, and data augmentation. The raw EEG and EMG data were first retrieved and structured based on subject and gesture-specific directories, ensuring proper alignment of multi-trial recordings.

1) ARTIFACT REMOVAL AND FILTERING

To remove power-line interference at 60 Hz, a second-order IIR notch filter with a quality factor of 30 was applied using a zero-phase forward-backward filtering approach. The notch filter is designed using the transfer function:

$$H(s) = \frac{s^2 + \omega_0^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2} \quad (1)$$

where ω_0 represents the notch frequency (60 Hz), and Q is the quality factor. Applying a zero-phase forward-backward filtering technique preserves the integrity of physiological signals by minimizing phase distortion.

Following notch filtering, a fourth-order Butterworth bandpass filter was applied to retain relevant frequency components. The bandpass filtering operation ensures that only the physiologically relevant EEG and EMG signals are preserved while attenuating unwanted noise and artifacts. The transfer function of an n^{th} -order Butterworth filter is given by:

$$H(s) = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2n}} \quad (2)$$

where ω_c denotes the cutoff frequency, and n represents the filter order. The bandpass filtering parameters were configured to pass EEG and EMG frequencies within 5 Hz to 50 Hz. A second-order section (SOS) implementation enhanced numerical stability and prevented signal distortion.

2) TEMPORAL SYNCHRONIZATION AND SEGMENTATION

Given the inherent differences in EEG and EMG sampling rates, a Fourier-based resampling technique was employed to standardize both modalities to a uniform frequency of 200 Hz, ensuring temporal alignment. The processed signals were then segmented into 1000 ms windows with an 80% overlap. This overlap was empirically chosen to maximize temporal continuity, preserve sufficient variability in gesture transitions, and ensure optimal segmentation for real-time analysis. The consistent overlap across datasets was necessary to mitigate intra-subject signal variability while maintaining a balance between computational efficiency and feature diversity. Each segmented window was then labeled according to its corresponding gesture category.

3) NORMALIZATION AND DATA AUGMENTATION

To mitigate inter-subject variability, Min-Max scaling was applied to normalize EEG and EMG signals within the range [0,1], enhancing model stability. The Min-Max normalization transformation is given by:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (3)$$

where $X_{\min}$ and $X_{\max}$ denote the minimum and maximum values of each channel, respectively. This normalization step ensured that all signals were uniformly scaled, preventing disparities in feature magnitude across subjects.

TABLE 1. EEG-EMG gesture datasets.

Dataset	Dataset-1	Dataset-2
Type	Scalp EEG and sEMG	Scalp EEG and sEMG
Subjects	33 volunteers	11 volunteers (7 males, 3 females, 1 left-handed male)
Gestures	7 types (G1: large diameter grasp, etc.)	Motor Imagery (MI) and grasp tasks
Repetitions	6 per gesture	6 per gesture
Electrodes	8 for EEG, 8 for EMG	8 for EEG (F3, F4, C3, CZ, C4, P3, PZ, P4), 8 for EMG
Sampling Rate	EEG: 250 Hz, EMG: 200 Hz	EEG: 250 Hz, EMG: 200 Hz
Total Time of All Recordings	23.1 hours	Not specified
Training Samples	5296	5276
Test Samples	1324	1304

TABLE 2. Description of hand gestures.

Gesture	Description
G1	Large diameter grasp
G2	Medium diameter grasp
G3	Three-finger sphere grasp
G4	Prismatic pinch grasp
G5	Power grasp
G6	Cut grasp
G7	Open hand

To improve generalization of the model, various data augmentation techniques were applied to the EEG and EMG signals. Gaussian noise injection was utilized by adding zero-mean Gaussian noise with a standard deviation of 0.01, simulating real-world signal variations and improving noise resilience. Amplitude scaling was implemented by multiplying the signal by a factor of 1.1, introducing variability in muscle contraction intensity and enabling the model to better adapt to inter-subject differences. Additionally, time shifting was performed by shifting EEG and EMG signals by five samples, addressing minor misalignments in sensor placement and improving temporal variability tolerance. To account for potential hardware fluctuations, frequency perturbation was introduced by slightly modifying the signal frequency, enhancing robustness against calibration inconsistencies. Synthetic sample generation was applied by creating new samples through linear combinations of existing data, thereby expanding training diversity and mitigating overfitting. These augmentation strategies collectively enhanced model generalization, ensuring robust performance across different subjects and environmental conditions, ultimately contributing to the reliability and accuracy of gesture classification. The dataset was shuffled before being fed into the model to prevent sequential biases. This preprocessing pipeline, illustrated in Figure 1, ensures that EEG and EMG signals are optimally conditioned for feature extraction and classification, enhancing the reliability of the NeuroFusion-Trans model.

4) TEMPORAL SYNCHRONIZATION

To address the temporal misalignment between EEG and EMG signals due to differences in their signal generation processes and sampling rates, we employ a *sliding window cross-correlation* approach for dynamic synchronization. Given an EEG signal $x(t)$ and an EMG signal $y(t)$, we compute the cross-correlation function:

$$R_{xy}(\tau) = \sum_t x(t)y(t + \tau) \quad (4)$$

where $R_{xy}(\tau)$ measures the similarity between EEG and EMG signals at different time shifts τ . The optimal lag τ^* , corresponding to the highest correlation, is determined as:

$$\tau^* = \arg \max_{\tau} R_{xy}(\tau) \quad (5)$$

ensuring optimal alignment between EEG and EMG within each processing window.

The synchronization procedure follows these steps: First, EEG is resampled using Fourier-based interpolation to match the length of EMG. Next, both signals are segmented into overlapping windows, ensuring local temporal relationships are preserved. Within each window, EEG and EMG signals are normalized to zero mean and unit variance:

$$x'(t) = \frac{x(t) - \mu_x}{\sigma_x}, \quad y'(t) = \frac{y(t) - \mu_y}{\sigma_y} \quad (6)$$

where μ_x, σ_x and μ_y, σ_y denote the mean and standard deviation of EEG and EMG, respectively.

For each window, the cross-correlation function is computed, and the time shift τ^* corresponding to the peak correlation is identified. The EEG window is then shifted by τ^* using circular shifting to synchronize it with EMG. If $\tau^* > 0$, EEG is delayed; otherwise, EMG is adjusted. The synchronization performance is quantified using an average synchronization score:

$$S_{\text{sync}} = \frac{1}{N} \sum_{i=1}^N \max R_{xy}^{(i)} \quad (7)$$

where N represents the total number of windows.

To reduce the computational complexity of sliding window cross-correlation ($O(N^2)$), we implemented a Fast Fourier Transform (FFT)-based approach, reducing the complexity

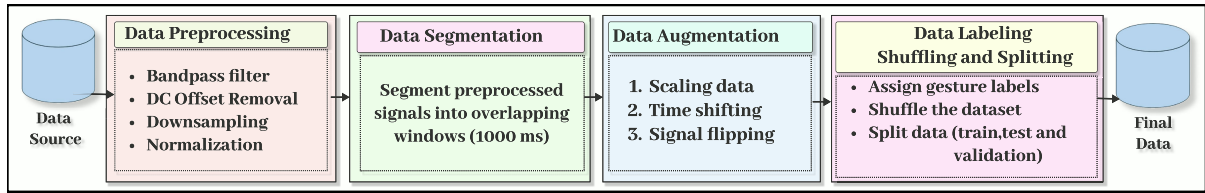


FIGURE 1. Overview of the EEG-EMG data processing pipeline: segmentation, labeling, shuffling, and splitting.

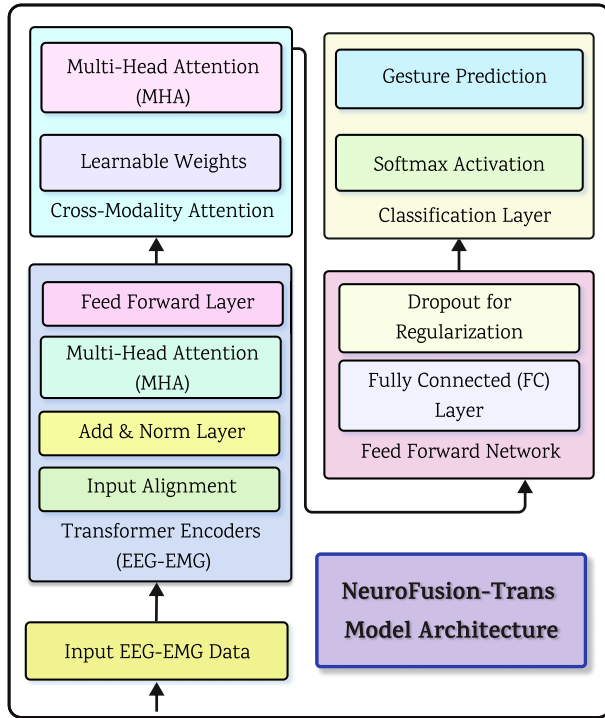


FIGURE 2. NeuroFusion-Trans model architecture.

to $O(N \log N)$ per window. This optimization significantly accelerates real-time synchronization, adding only approximately 0.0068 ms of computation per window during inference. Alternative methods such as Dynamic Time Warping (DTW) and its fast variants were also considered. However, DTW-based approaches typically introduce even higher computational costs for fine-grained alignment, making FFT-optimized cross-correlation a more practical choice for real-time EEG-EMG synchronization.

C. NEUROFUSION-TRANS MODEL ARCHITECTURE

The proposed NeuroFusion-Trans model addresses critical research gaps in cross-modality attention, temporal synchronization, and interactive online adaptation for EEG-EMG hybrid systems. Figure 2 illustrates the model architecture, detailing the input alignment, dual transformer encoders, cross-modality attention mechanism, and classification layers. Figure 3 outlines the complete workflow, highlighting the synchronization of EEG-EMG signals, modality-

specific encoding, dynamic feature fusion, online adaptation, and final gesture classification. Together, these figures provide a comprehensive view of how NeuroFusion-Trans dynamically fuses multimodal information and adapts to user-specific variations over time. The detailed training procedure and online adaptation strategy are further described in Algorithm 1.

The architectural details of the NeuroFusion-Trans model, including the hyperparameter configurations used to train the model, such as the learning rate, batch size, number of transformer layers, attention heads, and dropout rates, are outlined in Table 3. These design choices ensure the model effectively captures temporal dependencies and inter-modality relationships while maintaining computational efficiency for real-time applications. A grid search was utilized to select the optimal hyperparameter values, ensuring the model's performance generality.

1) MULTIMODAL SIGNAL ALIGNMENT

EEG and EMG signals are initially preprocessed using a sliding window cross-correlation technique to dynamically adjust for temporal misalignment. This ensures that both signals are synchronized before feature extraction. After this preprocessing step, a learnable transformation is applied to project EEG and EMG signals into a common latent space:

$$\begin{aligned} X_{\text{transformed}}^{\text{EEG}} &= W^{\text{EEG}} X^{\text{EEG}} + b^{\text{EEG}}, \\ X_{\text{transformed}}^{\text{EMG}} &= W^{\text{EMG}} X^{\text{EMG}} + b^{\text{EMG}} \end{aligned} \quad (8)$$

where W^{EEG} and W^{EMG} are trainable weight matrices, and b^{EEG} and b^{EMG} are bias terms. This transformation further refines the features, ensuring compatibility between the two modalities for effective fusion.

2) TRANSFORMER-BASED FEATURE ENCODING

The transformed EEG and EMG features are processed independently through two Transformer Encoder networks to extract intra-modal temporal dependencies:

$$\begin{aligned} H^{\text{EEG}} &= \text{TransformerEncoder}(X_{\text{transformed}}^{\text{EEG}}), \\ H^{\text{EMG}} &= \text{TransformerEncoder}(X_{\text{transformed}}^{\text{EMG}}) \end{aligned} \quad (9)$$

Each Transformer encoder employs multi-head self-attention (MHSA) to capture long-range dependencies:

$$\text{MHSA}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (10)$$

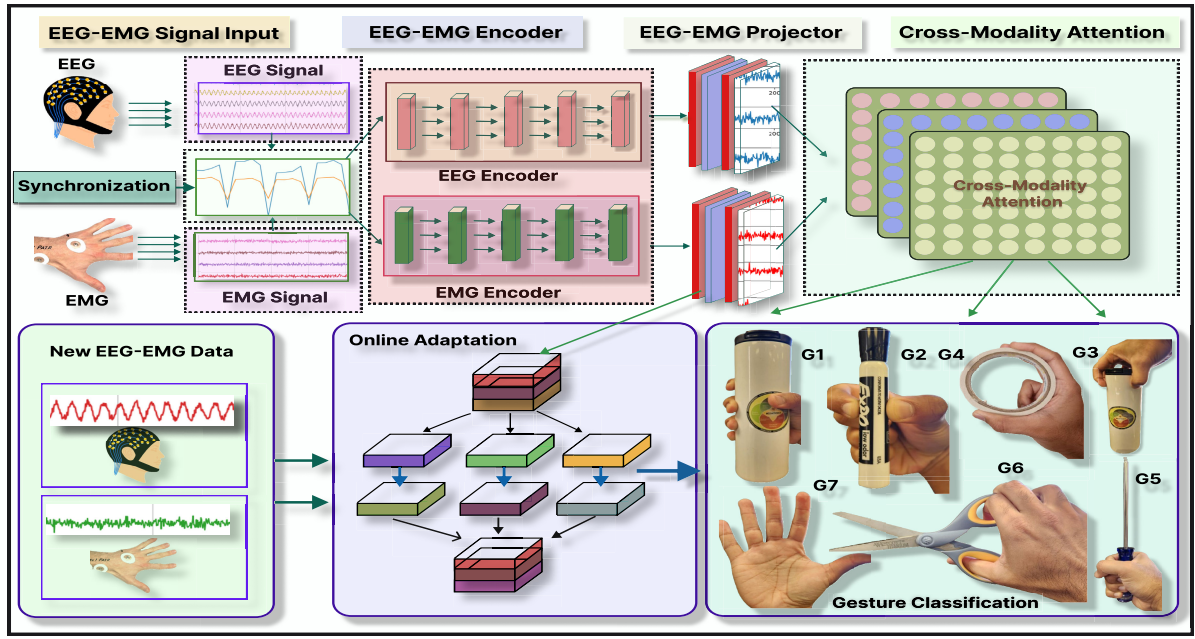


FIGURE 3. The workflow of the proposed NeuroFusion-Trans model that highlights the multimodal approach for real-time user intent recognition using EEG and EMG data.

where Q, K, V represent the query, key, and value matrices, and d is the embedding dimension. This mechanism allows the model to emphasize task-relevant information while mitigating the effects of noise in both EEG and EMG signals.

3) CROSS-MODALITY ATTENTION MECHANISM

To enable dynamic fusion of EEG and EMG features, a cross-modality attention mechanism is introduced. This mechanism adaptively weighs the importance of each modality, allowing the model to adjust dynamically based on signal quality and task relevance:

$$H^{fusion} = \alpha_{EEG}H^{EEG} + \alpha_{EMG}H^{EMG} \quad (11)$$

where $\alpha_{EEG}, \alpha_{EMG}$ are trainable parameters optimized during training. These weights, initialized with a same preference for EEG and EMG, enable the model to prioritize the most reliable modality under varying conditions. Further enhancement of inter-modality interactions is achieved through multi-head cross-attention. The cross-modality attention mechanism is initialized with a preference for EEG and EMG signals based on prior neurophysiological reliability. These modality weights are treated as trainable parameters and dynamically updated during training through backpropagation to optimize gesture classification performance. Furthermore, during online adaptation, the model fine-tunes these weights based on user-specific and session-specific variations, ensuring robustness against unstable or evolving signal quality conditions.

$$H^{cross} = \text{MHSA}(H^{fusion}, H^{fusion}, H^{fusion}) \quad (12)$$

This mechanism facilitates effective information exchange between EEG and EMG features, leveraging their complementary nature for enhanced intent recognition.

4) CLASSIFICATION AND ONLINE ADAPTATION

The fused representation undergoes temporal pooling before being processed by a fully connected layer for classification:

$$Y_{pred} = \text{softmax}(W_{fc} \cdot \text{Mean}(H^{cross}) + b_{fc}) \quad (13)$$

where W_{fc} and b_{fc} are trainable parameters of the classification layer, and $\text{Mean}(H^{cross})$ represents temporal averaging.

a: ONLINE ADAPTATION

To enhance real-time adaptability, the model incorporates an incremental learning mechanism that fine-tunes parameters using a small fraction of new data. This enables dynamic adjustments to user-specific variations, such as muscle fatigue, environmental fluctuations, or session-specific shifts, ensuring sustained accuracy over time. The adaptation process integrates a replay buffer to retain a subset of past training data, mitigating catastrophic forgetting by reinforcing previously learned patterns. To maintain stability, the total loss function includes a regularization term, preventing excessive deviation from prior knowledge:

$$L_{total} = L_{CE}(Y_{pred}, Y_{true}) + \lambda \|\theta - \theta_{prev}\|^2 \quad (14)$$

where L_{CE} is the categorical cross-entropy loss, and θ_{prev} represents the pre-adaptation parameters.

b: REAL-TIME ADAPTATION SETUP

During online operation, the NeuroFusion-Trans model adapts in real time by leveraging a cyclic learning rate schedule (triangular2 policy) to ensure controlled parameter updates. The adaptation process uses mini-batches of newly acquired user data along with a replay of past training samples to stabilize learning and prevent catastrophic forgetting. Gradient clipping and L2 regularization are applied during each update step to maintain model stability, and an early stopping mechanism halts adaptation if validation loss increases, thereby avoiding model drift. This lightweight adaptation strategy enables fast real-time fine-tuning while maintaining high classification accuracy across evolving user conditions.

5) ADAPTIVE ONLINE LEARNING FOR ROBUST PERFORMANCE

To further optimize online adaptation, NeuroFusion-Trans employs a cyclic learning rate scheduler in a triangular2 mode, which oscillates between a base and peak learning rate, ensuring controlled parameter updates. The adaptation process operates in mini-batches, utilizing only 30% of newly encountered data compared to validation data per cycle to enhance adaptability while preventing overfitting. To ensure stable updates, the model applies L2 regularization (weight decay) and gradient clipping, preventing excessive parameter adjustments. Additionally, an early stopping mechanism monitors validation loss and halts adaptation if an increase is detected, preserving model integrity. By integrating these techniques, NeuroFusion-Trans dynamically refines its classification performance, necessary for assistive robotics and neurorehabilitation applications.

6) MODEL HYPERPARAMETER CONFIGURATION

The hyperparameters used in NeuroFusion-Trans are optimized through a grid search strategy to ensure robust performance across subjects and sessions in Table 3.

These configurations ensure an optimal balance between learning efficiency, real-time adaptability, and generalization, which makes the model well-suited for assistive robotics applications. The cross-modality attention mechanism adaptively assigns weights to EEG and EMG features, initially favoring EEG ($\alpha_{EEG} = 0.7$, $\alpha_{EMG} = 0.3$), but dynamically optimizing these values during training.

The model fine-tunes using 400 data points for online adaptation, employing low learning rates, replay memory, and regularization constraints to prevent catastrophic forgetting while adapting to user-specific variations.

D. BASELINE AND COMPARATIVE MODELS

To evaluate the effectiveness of the proposed NeuroFusion-Trans model, we compare its performance against several established deep learning architectures commonly used in EEG-EMG-based gesture recognition. These models, which

TABLE 3. Hyperparameter configuration for NeuroFusion-Trans.

Hyperparameter	Value
General Model Parameters	
Input Dimension	Variable
Hidden Dimension	256
Transformer Layers	2
Attention Heads	4
Dropout Rate	0.5
Batch Size	32
Learning Rate	0.0005
Weight Decay	0.01
Epochs	15
Online Adaptation Parameters	
Online Adaptation Data	400
Online Learning Cycles	5
Online Batch Size	16
Online Learning Rate	0.00001
Learning Rate Scheduler	CyclicLR (Triangular2)
Gradient Clipping	1.0
Regularization Coefficient λ	0.01
Loss Function	Cross-Entropy Loss + Regularization

span convolutional, recurrent, and hybrid architectures, serve as strong baselines for benchmarking the proposed approach.

1) CNNLSTMNET

A hybrid deep learning architecture that integrates Convolutional Neural Networks (CNNs) for feature extraction and LSTM networks for sequential modeling. The CNN layers extract spatial features from EEG-EMG signals, while the LSTM layers capture temporal dependencies. The classification output is computed as:

$$h_t = \sigma(W_f * x_t + U_f * h_{t-1} + b_f), \quad (15)$$

where h_t is the hidden state at time step t , and W_f , U_f , and b_f are trainable parameters.

2) SHALLOWCONVNET

A compact CNN-based architecture designed for EEG classification. It employs shallow convolutional layers followed by pooling operations to capture low-level spatial patterns. A softmax classifier is used to predict the gesture class:

$$p(y = k|x) = \frac{e^{z_k}}{\sum_j e^{z_j}}, \quad (16)$$

where z_k represents the activation of the output layer corresponding to class k .

3) LSTMNET

A sequential model based purely on LSTM networks, designed to capture long-term dependencies in EEG-EMG signals. The update equations for the LSTM cell are

Algorithm 1 EEG-EMG Transformer Training and Online Adaptation

EEG data $\mathbf{X}_{\text{EEG}} \in \mathbb{R}^{N_{\text{EEG}} \times D_{\text{EEG}}}$, EMG data $\mathbf{X}_{\text{EMG}} \in \mathbb{R}^{N_{\text{EMG}} \times D_{\text{EMG}}}$, labels $\mathbf{y} \in \{1, \dots, C\}^N$, model M_θ , number of folds k , batch size B , learning rate α , online data proportion p_{online} . Trained model M_{θ^*} , metrics: Precision ($\mathcal{P}$), Recall ($\mathcal{R}$), F1 ($\mathcal{F}_1$), Accuracy ($\mathcal{A}$).

Preprocessing: Synchronize EEG and EMG data using cross-correlation:

$$\mathbf{X}_{\text{aligned}} = \text{Sync}(\mathbf{X}_{\text{EEG}}, \mathbf{X}_{\text{EMG}})$$

Encode labels $\mathbf{y}$.

Cross-Validation: Split data into k folds.

$f \leftarrow 1$ k Initialize M_θ with weight decay and cyclic learning rate.

Training: Optimize model parameters:

$$\theta_t = \theta_{t-1} - \alpha \nabla_{\theta} \mathcal{L}(M_\theta(\mathbf{X}_{\text{train}}^f), \mathbf{y}_{\text{train}}^f)$$

Apply gradient clipping and early stopping if validation loss $\mathcal{L}_{\text{val}}$ does not improve for P epochs.

Evaluate: Compute metrics $\mathcal{P}$, $\mathcal{R}$, $\mathcal{F}_1$, $\mathcal{A}$, Balanced Accuracy ($\mathcal{BA}$), Cohen's Kappa ($\mathcal{K}$), and Matthews Correlation Coefficient ($\mathcal{MCC}$).

Online Adaptation: Sample online data:

$$\mathbf{X}_{\text{online}}, \mathbf{y}_{\text{online}} = \text{Sample}(\mathbf{X}_{\text{val}}, p_{\text{online}})$$

Create replay buffer with stored training data:

$$\mathbf{X}_{\text{buffer}} = \text{ReplayBuffer}(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$$

Update model with regularized online adaptation:

$$\theta_t^{\text{online}} = \theta_{t-1}^{\text{online}} - \alpha_{\text{online}} \nabla_{\theta} \mathcal{L}(M_\theta(\mathbf{X}_{\text{online}}, \mathbf{X}_{\text{buffer}}), \mathbf{y}_{\text{online}}, \mathbf{y}_{\text{buffer}})$$

Evaluate Post-Adaptation: Recompute metrics.

Store Best Model:

$$M_{\theta^*} = \text{argmax}_f(\mathcal{A}_{\text{val}}^f)$$

Final Output: Compute mean metrics across k folds:

$$\mathbb{E}[\mathcal{P}], \mathbb{E}[\mathcal{R}], \mathbb{E}[\mathcal{F}_1], \mathbb{E}[\mathcal{A}], \mathbb{E}[\mathcal{BA}], \mathbb{E}[\mathcal{K}], \mathbb{E}[\mathcal{MCC}]$$

Computational Analysis: Measure inference time, model size, and attention weights. Visualize confusion matrix, ROC, precision-recall curves, and modality contributions.

defined as:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), \quad (17)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i), \quad (18)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o), \quad (19)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tanh(W_c x_t + U_c h_{t-1} + b_c), \quad (20)$$

where f_t , i_t , and o_t are the forget, input, and output gates, respectively.

4) VAE (VARIATIONAL AUTOENCODER)

A generative model that learns a latent representation of EEG-EMG signals for classification and reconstruction. The VAE objective function consists of a reconstruction loss and a Kullback-Leibler (KL) divergence regularization term:

$$\mathcal{L} = \mathbb{E}_{q(z|x)}[\log p(x|z)] - D_{KL}(q(z|x)||p(z)), \quad (21)$$

where $q(z|x)$ is the encoder's latent distribution, and $p(z)$ is the assumed prior.

5) TEMPORAL CONVOLUTIONAL NETWORK (TCN)

A fully convolutional alternative to recurrent models, leveraging dilated convolutions to capture long-range temporal dependencies efficiently. The output at time step t is computed as:

$$y_t = \sum_{i=0}^{k-1} w_i x_{t-d \cdot i}, \quad (22)$$

where w_i are the convolutional filter weights, d is the dilation factor, and k is the filter size.

These models represent a diverse set of architectures, ranging from convolutional and recurrent networks to hybrid and generative models. Their inclusion ensures a comprehensive evaluation of the proposed NeuroFusion-Trans framework in terms of classification accuracy, temporal synchronization, and adaptability.

E. TRAINING AND EXPERIMENTAL SETUP

The training phase employs k-fold cross-validation for a rigorous performance assessment, using Adam optimization and early stopping to prevent overfitting. Metrics include accuracy, precision, recall, F1-score, balanced accuracy, MCC, and Cohen's Kappa. An online adaptation mechanism incrementally refines model parameters, mitigating catastrophic forgetting and enhancing adaptability. Real-time optimizations ensure low-latency predictions, making the model suitable for assistive robotics. Experiments are conducted in PyTorch on an NVIDIA RTX 3500 Ada GPU with batch size 32 and a dynamic learning rate scheduler. Training takes 2 minutes per epoch for Dataset 1 and 2.5 minutes for Dataset 2. Despite the computational demands of transformers, optimizations such as mixed precision training and efficient attention mechanisms ensure minimal inference latency, supporting interactive applications.

IV. RESULT AND DISCUSSION

The proposed NeuroFusion-Trans framework is evaluated using two publicly available EEG-EMG datasets focused on upper-limb gesture classification. To evaluate the performance of the model, key metrics such as accuracy, F1 score, Cohen's Kappa, and Matthew's correlation coefficient (MCC) are used. A comparative study is conducted against multiple state-of-the-art baseline models, along with ablation experiments to quantify the contribution of critical components: temporal resolution enhancement, cross-modality attention, and adaptive online learning. To examine classification variability, Figure 4 illustrates heat maps of Cohen's Kappa scores across gestures (G1–G7) and cross-validation folds for both datasets. These visual representations provide information on the robustness of the model, highlighting its consistency between gestures and different validation splits.

A. OVERALL PERFORMANCE ANALYSIS

Table 4 and Table 5 summarize the classification performance of NeuroFusion-Trans and baseline models on Dataset 1 and

TABLE 4. Comparison of model performance on EMG-EEG dataset for upper-Limb gesture classification (Dataset 1).

Model	Precision	Recall	F1-score	Accuracy	Cohen's Kappa	Balanced Accuracy	MCC
NeuroFusion-Trans	0.96	0.96	0.96	0.96	0.96	0.96	0.96
<i>Online Adaptation</i>	<i>+0.01</i>	<i>+0.01</i>	<i>+0.01</i>	<i>+0.01</i>	<i>+0.01</i>	<i>+0.012</i>	<i>+0.01</i>
ShallowConvNet	0.47	0.61	0.52	0.61	0.6120	0.54	0.5636
<i>Online Adaptation</i>	-0.18	-0.22	-0.18	-0.22	-0.22	-0.25	-0.23
LSTMNet	0.46	0.60	0.51	0.60	0.6060	0.54	0.5577
<i>Online Adaptation</i>	<i>+0.13</i>	-0.37	-0.34	-0.37	-0.37	-0.43	-0.42
GRUNet	0.47	0.60	0.52	0.60	0.6086	0.54	0.5608
<i>Online Adaptation</i>	-0.02	-0.37	-0.36	-0.37	-0.38	-0.44	-0.43
CNNLSTMNet	0.19	0.25	0.17	0.25	0.2581	0.13	0.1523
<i>Online Adaptation</i>	-0.16	-0.10	-0.13	-0.10	-0.11	-0.13	-0.15
VAE	0.48	0.60	0.51	0.60	0.6022	0.53	0.5541
<i>Online Adaptation</i>	-0.08	-0.36	-0.34	-0.36	-0.37	-0.43	-0.42
CNNGRU	0.86	0.79	0.82	0.79	0.81	0.75	0.75
<i>Online Adaptation</i>	-0.06	-0.20	-0.17	-0.20	-0.19	-0.23	-0.20
CNNRNN	0.29	0.19	0.18	0.19	0.20	0.07	0.09
<i>Online Adaptation</i>	-0.08	-0.10	-0.11	-0.10	-0.10	-0.05	-0.06
TCN	0.16	0.16	0.11	0.16	0.17	0.07	0.08
<i>Online Adaptation</i>	-0.04	-0.05	-0.02	-0.05	-0.05	-0.01	-0.01

TABLE 5. Comparison of model performance on 8-Channel EMG-EEG upper limb gesture data (Dataset 2).

Model	Precision	Recall	F1-score	Accuracy	Cohen's Kappa	Balanced Accuracy	MCC
NeuroFusion-Trans	0.95	0.95	0.95	0.95	0.94	0.96	0.95
<i>Online Adaptation</i>	<i>+0.01</i>	<i>+0.01</i>	<i>+0.01</i>	<i>+0.01</i>	<i>+0.011</i>	<i>+0.01</i>	<i>+0.01</i>
ShallowConvNet	0.52	0.62	0.55	0.62	0.6211	0.56	0.60
<i>Online Adaptation</i>	-0.09	-0.20	-0.23	-0.20	-0.20	-0.23	-0.23
LSTMNet	0.54	0.63	0.56	0.63	0.6260	0.56	0.61
<i>Online Adaptation</i>	<i>+0.04</i>	-0.02	<i>+0.02</i>	<i>+0.02</i>	<i>+0.02</i>	<i>+0.03</i>	<i>+0.02</i>
GRUNet	0.53	0.63	0.56	0.63	0.6250	0.56	0.61
<i>Online Adaptation</i>	-0.027	-0.018	<i>+0.24</i>	<i>+0.18</i>	<i>+0.18</i>	<i>+0.22</i>	<i>+0.17</i>
CNNLSTMNet	0.40	0.47	0.39	0.47	0.4691	0.38	0.42
<i>Online Adaptation</i>	-0.03	-0.08	-0.07	-0.08	-0.08	-0.10	-0.10
VAE	0.51	0.61	0.53	0.61	0.6085	0.54	0.58
<i>Online Adaptation</i>	<i>+0.00</i>	-0.08	-0.09	-0.08	-0.08	-0.09	-0.10
CNNGRU	0.54	0.52	0.48	0.52	0.50	0.41	0.44
<i>Online Adaptation</i>	-0.05	-0.09	-0.06	-0.09	-0.08	-0.05	-0.06
CNNRNN	0.26	0.32	0.26	0.32	0.33	0.17	0.20
<i>Online Adaptation</i>	-0.02	-0.09	-0.07	-0.09	-0.12	-0.08	-0.10
TCN	0.21	0.23	0.15	0.23	0.22	0.09	0.11
<i>Online Adaptation</i>	-0.04	-0.07	-0.04	-0.07	-0.07	-0.04	-0.04

Dataset 2, respectively. The proposed model achieves an accuracy of 97% and 96% on Dataset 1 and Dataset 2, outperforming all other models. The significant improvement in F1-score (0.97 and 0.95, respectively) highlights the generality and strength of the model in capturing both EEG and EMG modalities effectively. Figure 5 presents the performance metrics comparison for Dataset 1 and Dataset 2, including accuracy and F1-score before and after online adaptation, as well as pre- and post-synchronization results.

B. EEG-EMG SIGNAL VISUALIZATION AND GESTURE-SPECIFIC CLASSIFICATION METRICS

We plotted the raw signals corresponding to each gesture to establish a direct relationship between EEG-EMG signals and gesture recognition (G1–G7). These visualizations illustrate how specific EEG and EMG modalities patterns correspond to different motor gestures, highlighting variations in amplitude, activation timing, and signal dynamics.

Representative examples are shown in Figure 6. EEG signals exhibit distinct cortical activation patterns, while

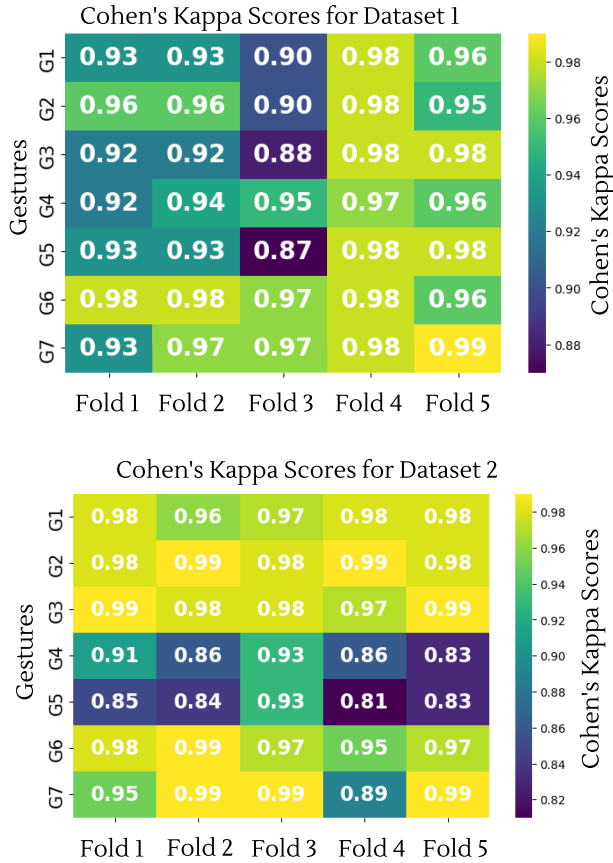


FIGURE 4. Heatmaps illustrating Cohen's Kappa scores across gestures (G1-G7) and cross-validation folds for two datasets.

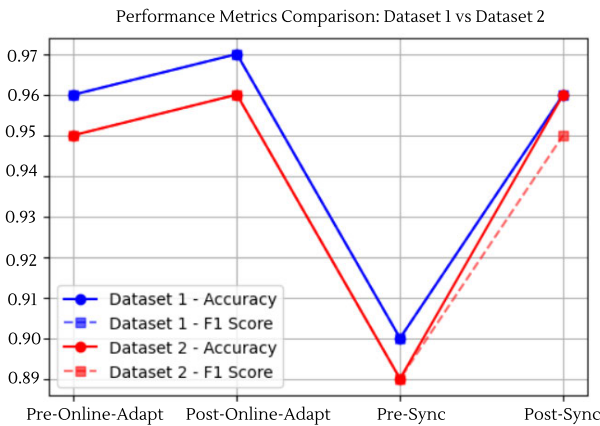


FIGURE 5. Performance metrics comparison for Dataset 1 and Dataset 2, illustrating accuracy and F1-score before and after online adaptation, as well as pre- and post-synchronization results.

EMG signals capture peripheral muscle activities during gesture execution, supporting the discriminative potential of multimodal signals for intent recognition. Gesture-specific precision, recall, and F1-score are detailed in Table 6. These results demonstrate that even using raw EEG and EMG

TABLE 6. Gesture-wise performance metrics based on EEG-EMG signals.

Gesture	Precision	Recall	F1-Score
G1	1.00	0.85	0.89
G2	0.90	1.00	0.90
G3	1.00	1.00	0.99
G4	0.95	0.99	0.91
G5	0.81	0.89	0.85
G6	1.00	0.99	1.00
G7	1.00	0.83	0.95

signals, distinct gesture classes can be reliably separated. This validates the discriminative capability of multimodal neurophysiological signals before applying deeper feature extraction and transformer-based modeling.

C. IMPACT OF TEMPORAL RESOLUTION ENHANCEMENT

One of the critical contributions of NeuroFusion-Trans is its ability to dynamically synchronize EEG and EMG signals, improving temporal resolution. Before applying the synchronization module, the accuracy for Dataset 1 and Dataset 2 was 90% and 89%, respectively. After incorporating the synchronization mechanism, the accuracy increased to 96% and 95%. These results emphasize the importance of precise temporal alignment in multimodal neurophysiological signal processing. Figure 7 illustrates the synchronization process, showing the original EEG signal and the synchronized EEG-EMG signals.

Figure 8 shows the temporal dynamics of gesture predictions for both datasets, illustrating how NeuroFusion-Trans adapts to real-time variations effectively and highlights the model's ability to handle dynamic variations and maintain consistent prediction accuracy over time.

D. EFFECTIVENESS OF CROSS-MODALITY ATTENTION MECHANISM

To evaluate the impact of the cross-modality attention mechanism, an ablation study was conducted by removing this module. The results indicate a significant performance drop, with accuracy reducing by 6.1%. This validates the effectiveness of cross-modal interaction modeling in leveraging interdependencies between EEG and EMG signals. The F1-score degradation further highlights the loss of discriminative information when EEG and EMG signals are not fused optimally. Figure 9 provides information on the representation of the latent space, demonstrating the importance of cross-modal interactions. The figure demonstrates the improved separation and clustering of latent features when the EEG and EMG signals are fused optimally, compared to the absence of the cross-modality attention module.

E. ADAPTIVE ONLINE LEARNING FOR ROBUST PERFORMANCE

The integration of adaptive online learning in NeuroFusion-Trans is pivotal for real-time personalization as it enables the model to dynamically adjust its parameters based on

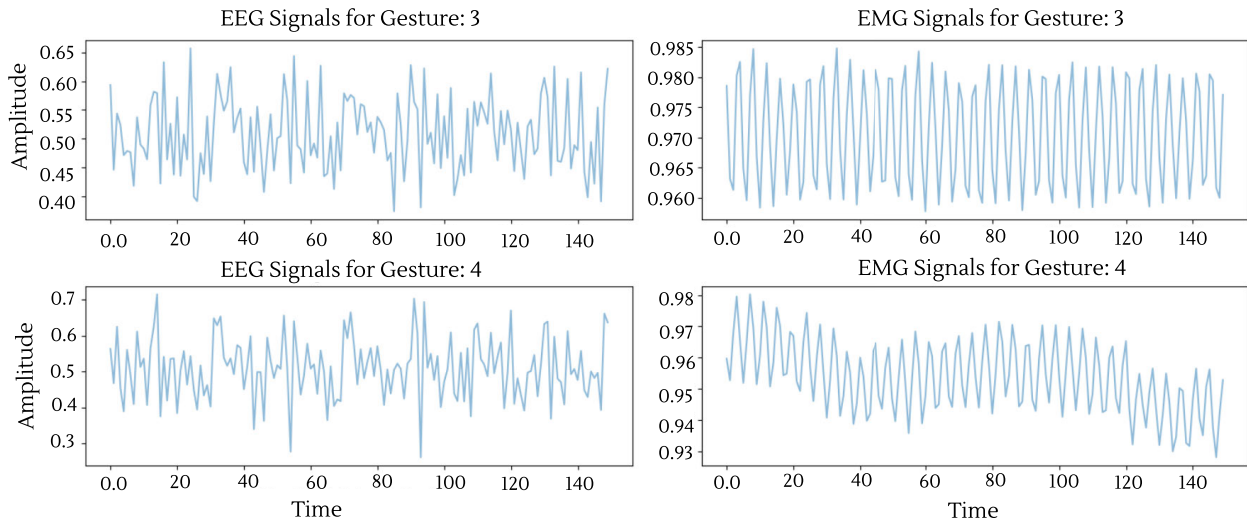


FIGURE 6. Gesture-specific EEG and EMG signal patterns for G3 and G4. Each plot shows characteristic neuromuscular activation patterns corresponding to distinct upper-limb gestures.

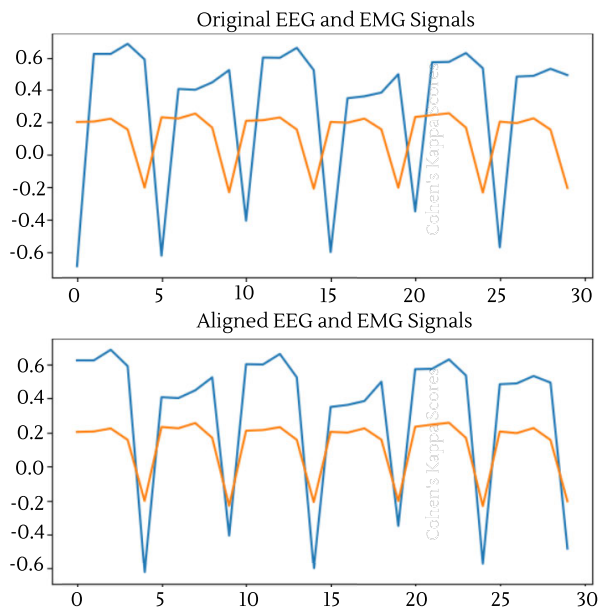


FIGURE 7. Illustration of the synchronization process, showing the original EEG signal and the synchronized EEG-EMG signals.

user-specific variations. The impact of online adaptation is evident through performance improvements across all evaluation metrics. For Dataset 1, online adaptation increased accuracy from 96% to 97%, while for Dataset 2, accuracy improved from 95% to 96%. Additionally, the F1-score exhibited a 1% improvement across both datasets, highlighting the effectiveness of continuous learning in reinforcing model robustness. Beyond overall performance gains, online adaptation significantly reduced misclassification rates for

gestures G4 and G5, which were previously more challenging to classify due to their overlapping signal patterns.

Figures 11, 12 and 10 present the ROC curve, precision-recall curve, and learning rate progression for both datasets, illustrating model performance across all gesture classes. Notably, baseline models such as CNN-LSTM and GRUNet struggled to maintain stable performance post-adaptation, with some even experiencing degradation. This underscores that NeuroFusion-Trans effectively integrates online adaptation to enhance generalization while mitigating overfitting, ensuring sustained and adaptive learning over time.

F. COMPARISON WITH BASELINE MODELS

The proposed model is benchmarked against multiple baselines, including CNN-LSTM, GRUNet, LSTMNet, and ShallowConvNet. The results show that these conventional models lack effective fusion strategies for EEG and EMG modalities, leading to lower classification performance. CNN-LSTM, which serves as the baseline model, achieves an accuracy of only 19% on Dataset 1 and 40% on Dataset 2, indicating that simple deep learning architectures struggle to capture cross-modal dependencies effectively. VAE and CNNGRU models performed relatively better than CNN-LSTM. Still, they lagged behind NeuroFusion-Trans, particularly in Cohen's Kappa and Balanced Accuracy, confirming the necessity of advanced feature fusion techniques. Figures 13 illustrates the confusion matrices for both datasets, highlighting the improved classification performance of NeuroFusion-Trans.

G. COMPUTATIONAL COMPLEXITY AND DEPLOYMENT

To assess the real-time feasibility of NeuroFusion-Trans, we evaluated its computational cost, including FLOPs,

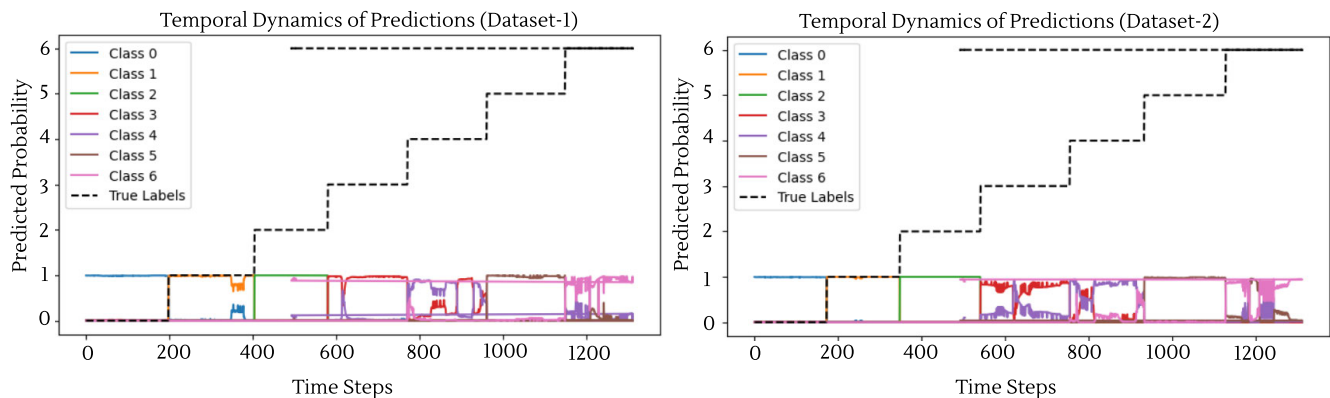


FIGURE 8. Temporal dynamics of gesture predictions for Dataset 1 and Dataset 2, illustrating the real-time adaptation capability of NeuroFusion-Trans.

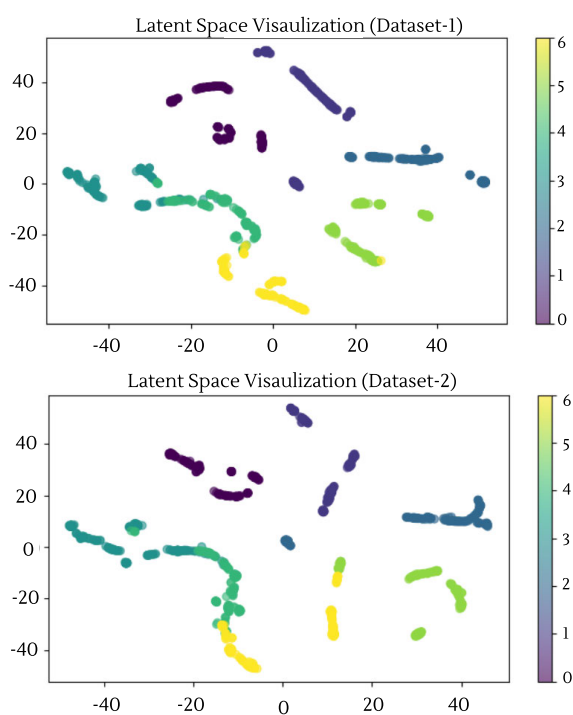


FIGURE 9. Latent space visualization illustrating the effect of the cross-modality attention mechanism.

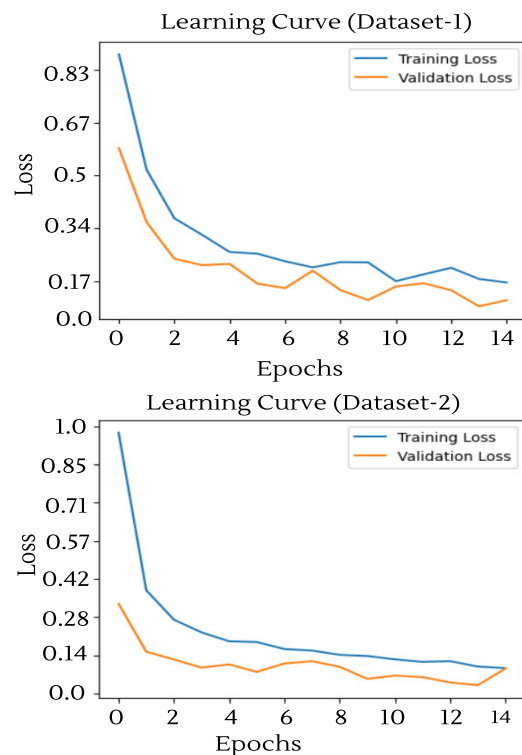


FIGURE 10. Learning rate progression for Dataset 1 and Dataset 2.

inference time, and latency. On an NVIDIA RTX 3500 Ada GPU, it achieved an average inference time of 0.0276 ms per sample, significantly outperforming a CNN-LSTM baseline of 0.4278 ms per sample, making it 15.5 $\times$ faster. This performance highlights the computational efficiency of multimodal fusion-based transformer for real-time assistive robotics. Despite the typical resource-intensive nature of transformers, we optimized inference speed through layer pruning, mixed precision training, and efficient attention mechanisms, ensuring practical feasibility. Future work will explore lightweight transformer variants (MobileViT, Tiny Transformer) to reduce computational overhead further and

enable deployment on edge devices, facilitating low-latency neurorehabilitation and human-machine interaction.

H. GENERALIZATION AND PRACTICAL IMPLICATIONS

The superior performance of NeuroFusion-Trans across two diverse EEG-EMG datasets demonstrates its ability to generalize effectively. The model achieves state-of-the-art accuracy and ensures robust synchronization and online adaptability, making it suitable for real-time assistive robotics applications. The ability to dynamically learn from user feedback and adjust to varying physiological conditions highlights its potential for applications in neurorehabilitation,

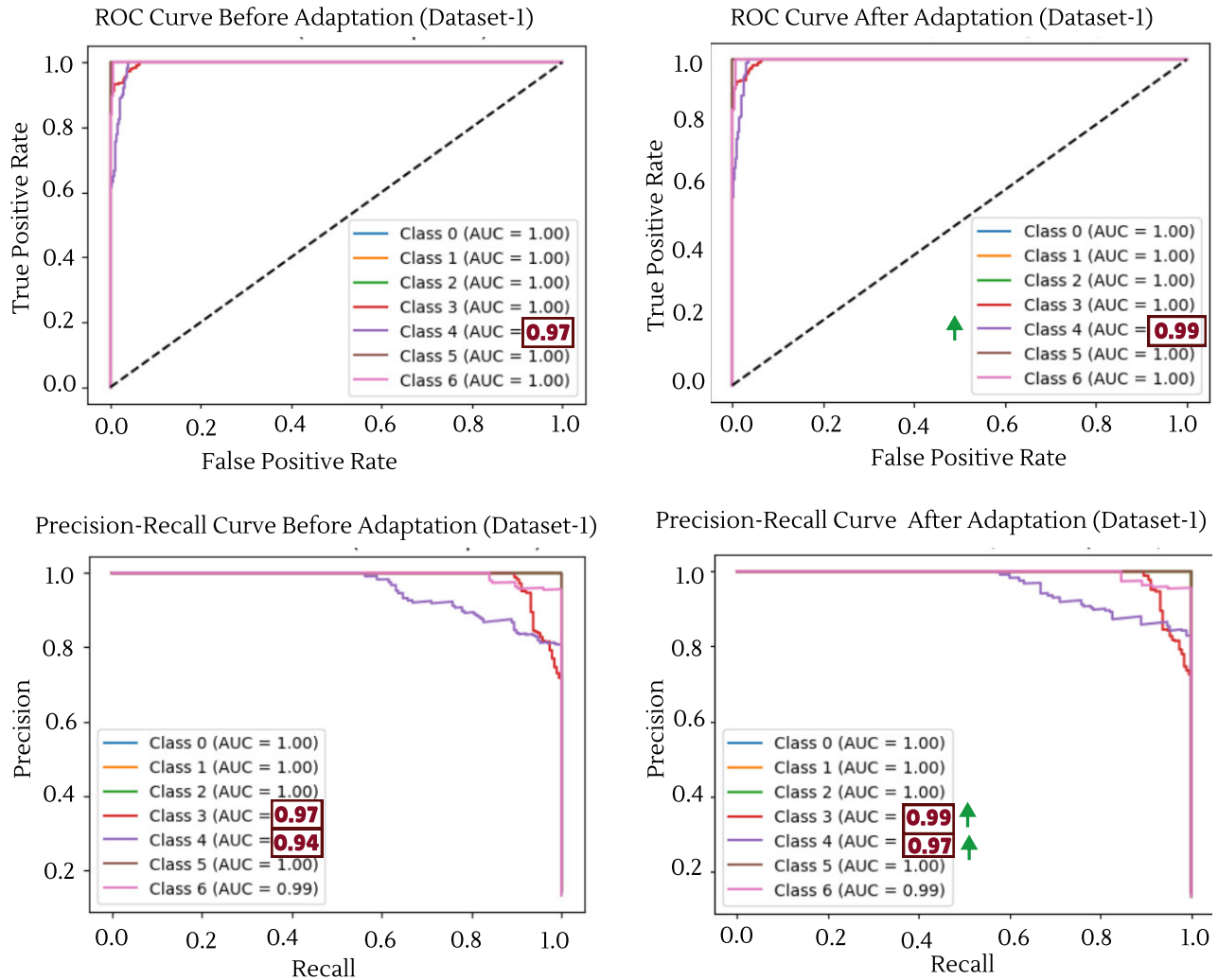


FIGURE 11. ROC curve, precision-recall curve for Dataset 1, illustrating the impact of adaptive online learning on class-wise performance.

prosthetic control, and brain-computer interfaces. Future research could explore expanding the model to incorporate additional biosignals such as electrooculography (EOG) or functional near-infrared spectroscopy (fNIRS) to further enhance intent recognition accuracy. The experimental results validate that NeuroFusion-Trans significantly enhances EEG-EMG gesture recognition by addressing key challenges in temporal alignment, cross-modality fusion, and adaptive learning. The proposed framework consistently outperforms state-of-the-art models, achieving the highest classification accuracy and synchronization scores. These findings highlight its potential for real-time human-machine interaction and neuroprosthetic control.

I. STATISTICAL ANALYSIS

Figure 13 shows the confusion matrices for NeuroFusion-Trans on both datasets, highlighting excellent classification performance with minimal misclassifications before and after the online adaptation. Gestures G1-G7 achieve consistently high scores, demonstrating the effectiveness of the

cross-modality attention mechanism in distinguishing subtle gesture variations. Figure 14 compares the mean accuracies of various models, with NeuroFusion-Trans achieving the highest accuracy and showing low-performance variability, as indicated by a narrow 95% confidence interval. This highlights the model's reliability for neurorehabilitation and assistive robotics. Slightly higher misclassification rates for gesture 4 and 5 are attributed to overlapping signal patterns, lower signal-to-noise ratios, or underrepresented gestures in training data.

J. ABLATION STUDY

To evaluate the significance of each component in NeuroFusion-Trans, we conducted an ablation study by systematically removing key modules and measuring their impact on performance. The results revealed that excluding the Cross-Modality Attention Mechanism led to a significant accuracy drop of around 6.1%, emphasizing the critical role of leveraging EEG-EMG interdependencies for improved intent recognition. Similarly, the disabling of the synchro-

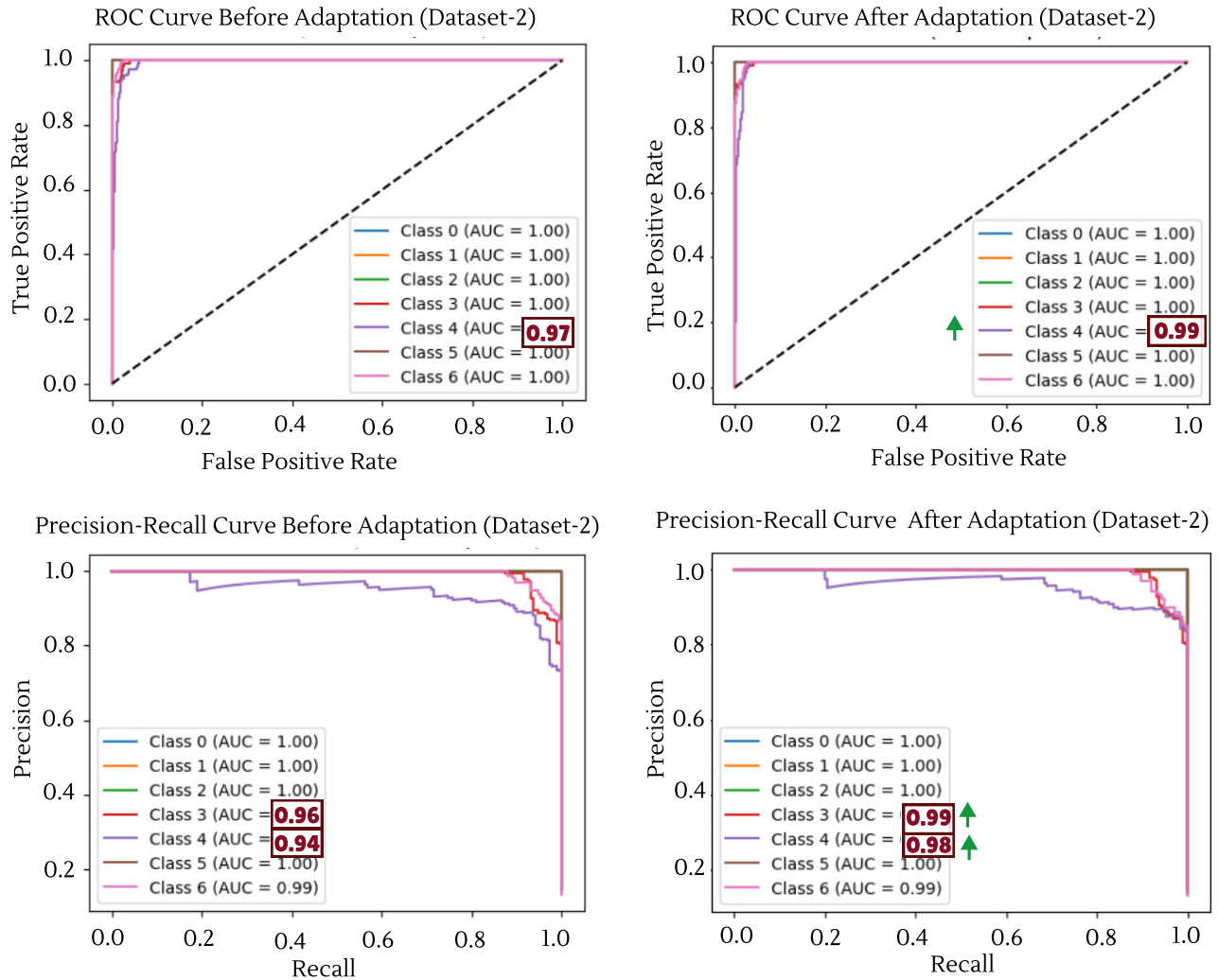


FIGURE 12. ROC curve, precision-recall curve for Dataset 2, illustrating the impact of adaptive online learning on class-wise performance.

nization mechanism resulted in a performance decline of around 6.7%, highlighting the necessity of synchronization for the modeling of general user intent. These findings validate the architectural choices of NeuroFusion-Trans and underscore the proposed mechanisms' individual contributions in enhancing the model's overall performance.

K. DISCUSSION

The proposed NeuroFusion-Trans framework effectively addresses EEG-EMG-based gesture recognition challenges by improving temporal resolution, integrating cross-modality attention, and incorporating adaptive online learning. These innovations significantly improve classification accuracy and robustness. The enhancement of temporal resolution ensures precise EEG-EMG synchronization, leading to a 6.7%. Cross-modality attention captures EEG-EMG dependencies, with ablation studies revealing a 6.1% accuracy drop when removed, underscoring its importance. The online adaptation performance analysis is shown in Figure 15. Adaptive online learning enhances personalization, improving accuracy,

F1-score, and kappa by 0.1% in both datasets, which is crucial for neurorehabilitation applications—for both datasets, accuracy, F1-score, and kappa increased by 0.1% after online adaptation. In Dataset-1, accuracy improved from 0.90 before synchronization to 0.96 after synchronization. Similarly, accuracy without the attention mechanism was 0.91, with a kappa value of 0.91, while incorporating attention improved the accuracy to 0.96 and kappa to 0.96. In Dataset-2, accuracy before synchronization was 0.89, increasing to 0.95 after synchronization. Without the attention mechanism, accuracy was 0.88 with a kappa value of 0.83, whereas using attention enhanced accuracy to 0.95 and kappa to 0.95. Compared to baseline models, NeuroFusion-Trans consistently outperforms traditional architectures like CNN-LSTM, GRU, and VAE, demonstrating superior multimodal integration and adaptability. It also achieves high EEG-EMG synchronization scores (0.7471 for Dataset 1 and 0.7687 for Dataset 2), ensuring smooth and responsive control mechanisms. NeuroFusion-Trans presents a robust solution for EEG-EMG gesture recognition,

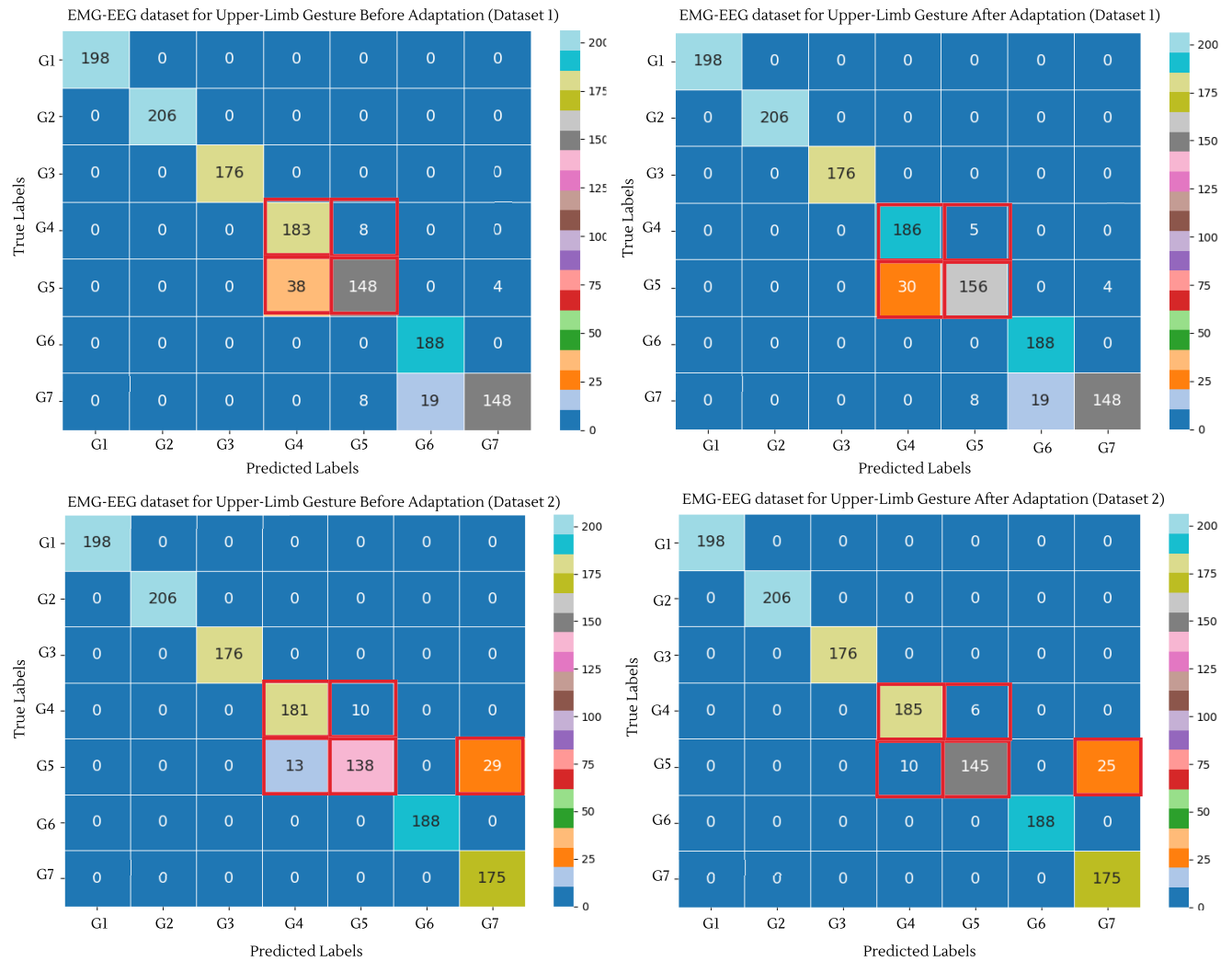


FIGURE 13. Confusion matrices for gesture classification with NeuroFusion-Trans before and after online adaptation for both datasets.

TABLE 7. State of the art in EEG-EMG classification for motor intent detection.

Ref	Model	Approach & Methodology	Performance Results
[11]	EEG-EMG FAConformer	A Conv-Transformer with frequency band attention and multi-scale fusion for EEG-EMG motor pattern recognition.	Achieves 62.5% accuracy on Multigrasp-MI.
[28]	Hierarchical EEG-EMG Fusion Model	Hierarchical deep learning model estimating EMG from EEG for movement and kinematics prediction.	Achieves 84.25% accuracy for finger movement prediction and an RMSE of 0.318 in kinematics estimation.
Proposed Model	NeuroFusion-Trans	A transformer-based EEG-EMG fusion model for real-time gesture recognition via temporal alignment, cross-modality attention, and adaptive learning.	Achieves 95% and 96% accuracy for gesture identification.

advancing assistive robotics and brain-computer interfaces through improved synchronization, attention mechanisms, and adaptive learning.

To the best of our knowledge, only two prior studies have explored EEG-EMG fusion models—FAConformer [11] and Hierarchical EEG-EMG Fusion [28]. As shown in Table 7, our proposed NeuroFusion-Trans surpasses these methods, achieving significantly higher classification accuracy

(95% and 96%). This improvement demonstrates the effectiveness of transformer-based cross-modality attention and adaptive learning for enhanced gesture recognition.

1) LIMITATIONS AND FUTURE CHALLENGES

While NeuroFusion-Trans achieves robust EEG-EMG fusion and superior gesture recognition performance, certain limitations remain. The model can struggle in scenarios with severe

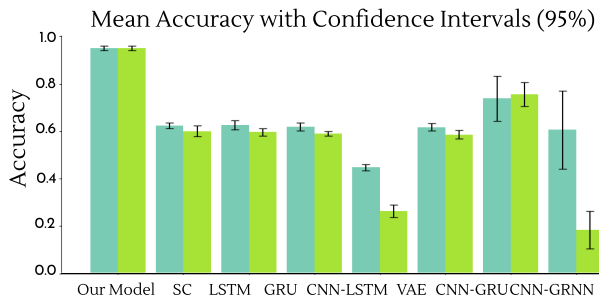


FIGURE 14. Comparison of mean accuracy across models for two datasets and error bars represent 95% confidence intervals.

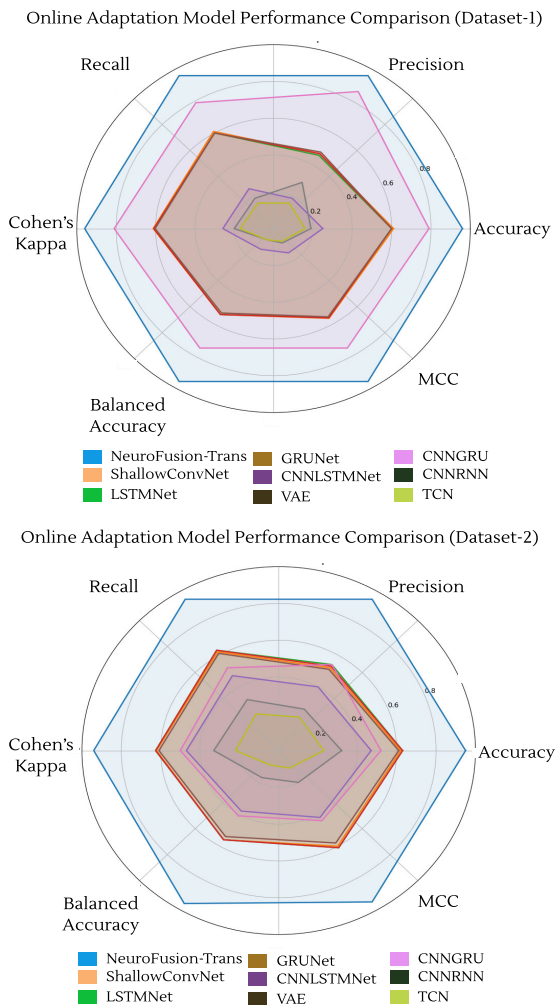


FIGURE 15. Online adaptation model performance comparison for both datasets.

signal degradation, such as excessive motion artifacts, strong electromagnetic interference, or poorly attached electrodes, where both EEG and EMG signals become highly noisy. Additionally, gestures involving subtle motor activities or overlapping muscle activations (e.g., G4 and G5) can

reduce cross-modality attention effectiveness, leading to misclassifications. Subject-specific factors, such as extreme physiological variability, muscle fatigue, or cognitive load variations, can also challenge the model's generalization. Although online adaptation mitigates some of these effects, rapid shifts in user conditions without sufficient adaptation cycles may still impact performance. Future improvements will explore more robust artifact removal, uncertainty modeling, and continual learning strategies to address these challenges.

V. CONCLUSIONS AND FUTURE DIRECTIONS

In this study, we introduced NeuroFusion-Trans, a transformer-based model for real-time user intent recognition from EEG-EMG signals. Our approach significantly enhances gesture recognition accuracy and robustness by addressing challenges such as temporal misalignment, weak cross-modality fusion, and limited adaptability. The cross-modality attention mechanism effectively integrates EEG and EMG signals, while adaptive online learning ensures continuous refinement to user-specific variations. Experimental results demonstrate state-of-the-art performance, with higher classification accuracy, Cohen's Kappa scores, and improved EEG-EMG synchronization compared to baseline models. Ablation studies validate the importance of key components, particularly attention-based fusion and adaptive learning, in improving classification performance.

Despite these advancements, future work will validate the model for neuromuscular impairment cases, explore self-supervised learning to reduce dependence on labeled data, and incorporate multi-scale temporal attention for enhanced feature extraction. Meta-learning and domain adaptation will be explored for rapid personalization and improved cross-user generalization. In addition, integrating real-time feedback loops will allow interactive refinement, making the model more adaptable for brain-controlled prosthetics and exoskeletons. The real-world deployment will be evaluated by integrating NeuroFusion-Trans into assistive robotic systems, assessing system latency, responsiveness, and usability in practical settings. NeuroFusion-Trans presents a significant step forward in EEG-EMG-based assistive technology, demonstrating strong potential for neuroprosthetics, rehabilitation, and brain-computer interfaces. Future research will enhance efficiency, adaptability, and real-world deployment, advancing human-machine interaction for individuals with motor impairments.

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TIPU SULTAN received the B.Sc. degree (Hons.) in mechatronics engineering from Kyungsuung University, South Korea, and the M.Sc. degree in data science from Lincoln Center, Fordham University, in April 2024. He is currently pursuing the Ph.D. degree in aerospace and mechanical engineering with Saint Louis University, where his research centers on developing transformer models to interpret user intent for robotic control. He is a Research Assistant with the Mechritm Laboratory, under the supervision of Dr. Madi Babaiaşl. He completed an internship as a Data Analyst with Authentic Brands Group. His expertise includes ML, tableau, power BI, and SQL database management, with over five years of experience, as a Data Analyst and a ML Researcher in diverse industries.



GUANGPING LIU (Student Member, IEEE) received the master's degree in mechanical engineering with a focus on mechatronics from Northeastern University, where she worked under the guidance of Dr. Seungmoon Song on self-pacing treadmill control. She is currently pursuing the Ph.D. degree in aerospace and mechanical engineering with Saint Louis University. She is also with the Mechritm Laboratory, under the supervision of Dr. Madi Babaiaşl, where her research focuses on assistive robotics that integrate vision-language-action models to develop user-friendly "WheelArms."



PASCAL SIKORSKI (Student Member, IEEE) is currently pursuing the B.S. degree in computer science and mathematics with Saint Louis University, with research interests in robotics, human-computer interaction, control theory, and game theory. In 2024, he was announced as a WAVE Fellow at California Institute of Technology, where he worked on control theory research, under Dr. Aaron Ames and Dr. Xiao Tan at the Caltech AMBER Laboratory. He is currently a Research Assistant with the Mechritm Laboratory, Saint Louis University, where he focuses on robotic control and human-robot interaction. Alongside his work in robotic control, he has explored vulnerabilities within visual SLAM and applications in the geospatial field. With a passion for automating robotics for everyday use, he hopes to pursue the Ph.D. degree and expand the accessibility of robotics for all.

SAMAH ALSHATHRI received the Bachelor of Computer Science and the Master of Computer Engineering degrees from King Saud University, Riyadh, Saudi Arabia, and the Ph.D. degree from the Department of Computer and Mathematics, Plymouth University, Plymouth, U.K. She is currently an Assistant Professor with the Department of Information Technology, College of Computer and Information Sciences, Princess Nourah bint Abdulrahman University (PNU), Riyadh. Her research interests include wireless networks, cloud computing, fog computing, the IoT, data mining, ML, text analytics, image classification, and DL. She was the Chair of the Network and Communication Department and participated in organizing many international conferences. She has authored or co-authored many articles published in well-known journals in the research field.



WALID EL-SHAFAI (Senior Member, IEEE) was born in Alexandria, Egypt. He received the B.Sc. degree (Hons.) in electronics and electrical communication engineering from the Faculty of Electronic Engineering (FEE), Menoufia University, Menouf, Egypt, in 2008, the M.Sc. degree from Egypt-Japan University of Science and Technology (E-JUST), in 2012, and the Ph.D. degree from FEE, Menoufia University, in 2019. From January 2021 to December 2024, he was a Researcher with the Security Engineering Laboratory (SEL), Prince Sultan University (PSU), Riyadh, Saudi Arabia. He is currently a Senior Researcher with the Automated Systems and Computing Laboratory (ASCL) and an Assistant Professor with the College of Computer Science and Information Systems, PSU. In addition, he is an Associate Professor with the Department of Electronics and Communication Engineering (ECE), Faculty of Electronic Engineering, Menoufia University. His research interests include wireless mobile and multimedia communication systems, image and video signal processing, efficient 2D/3D video coding and transmission, quality of service and experience, digital communication techniques, cognitive radio networks, adaptive filter design, 3D video watermarking, steganography, encryption, error resilience and concealment algorithms for video codecs (H.264/AVC, H.264/MVC, and H.265/HEVC), cognitive cryptography, medical image processing, speech processing, security algorithms, software-defined networks, the Internet of Things, FPGA implementations of signal processing algorithms and communication systems, cancellable biometrics, pattern recognition, image and video magnification, artificial intelligence applications in signal processing and communication systems, modulation identification and classification, image and video super-resolution and denoising, automated systems, cybersecurity applications, malware and ransomware detection, and DL for signal processing and communication systems. He is a dedicated reviewer for numerous international journals and conferences, contributing to the advancement of research in his areas of expertise.



MADI BABAIASL (Member, IEEE) received the Ph.D. degree from Washington State University, Pullman, WA, USA, with research focused on developing robotic steerable needles for surgical purposes. She is currently a Clare Boothe Luce Assistant Professor of robotics and autonomous systems and the Director of the Mecharithm Laboratory, Saint Louis University. Her current research interests include the intersection of robotics, ML, control theory, artificial intelligence, natural language processing, game development, psychology, physical therapy, and education. She is also a Professional Member of IEEE, IEEE Robotics and Automation Society, and IEEE Women in Engineering.

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